

Bobapedia: A Workshop on Archiving Living Flavor Experiences

Po-Yu Chen
National Taiwan University
Taipei, Taiwan
reach@poyuchentwmail.com

Eunsol Choi
MIT CSAIL
Cambridge, MA, USA
solch@mit.edu

Takuji Narumi
University of Tokyo
Tokyo, Japan
narumi@cyber.t.u-tokyo.ac.jp

Paul Pu Liang
MIT Media Lab, MIT EECS
Cambridge, MA, USA
ppliang@mit.edu

Jas Brooks
MIT CSAIL
Cambridge, MA, USA
jasb@mit.edu

Lung-Pan Cheng
National Taiwan University
Taipei, Taiwan
lung-pan.cheng@csie.ntu.edu.tw

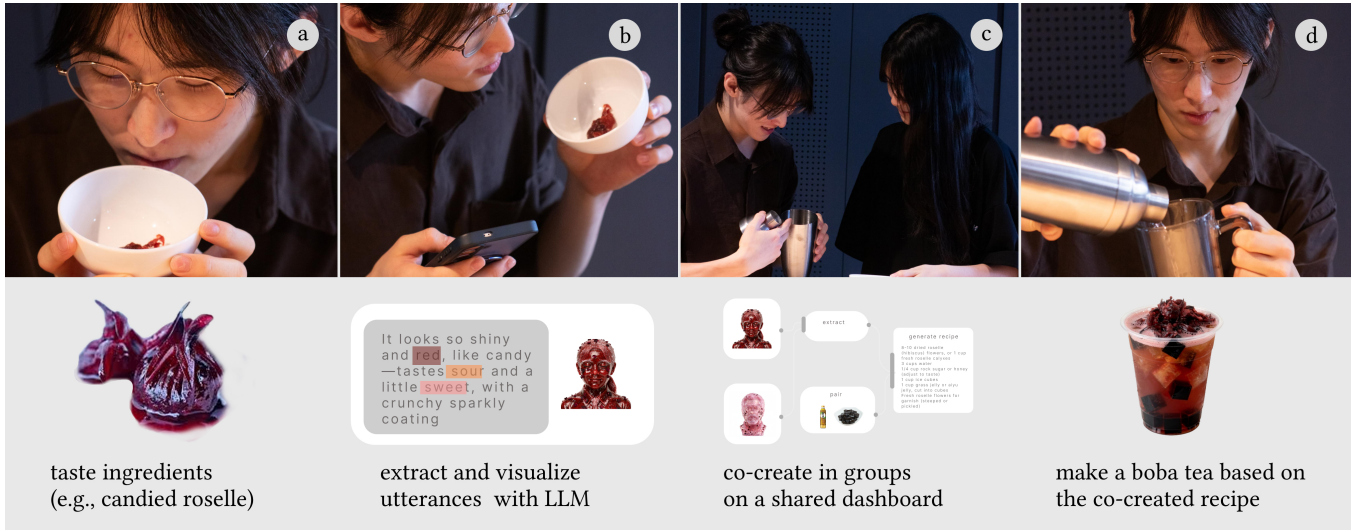


Figure 1: An example workflow in this workshop, illustrating how subjective flavor experiences are externalized, structured, and recomposed into a shared recipe. (a) Participants taste and smell the ingredients (e.g., candied roselle as an example). (b) Participants describe their flavor experience, which is recorded and visualized by an agentic system as an automatically generated knowledge graph. (c) Participants form groups to discuss and co-edit their flavor concepts on a shared dashboard. (d) Participants prepare a boba tea based on the group discussion and the resulting generated recipe.

Abstract

Flavor is an embodied, multimodal, and culturally situated design material. However, computational flavor systems often represent food through ingredients, chemical compounds, recipe corpora, or machine-generated recommendations, leaving limited space for people to document and iterate their subjective experiences. We propose *Bobapedia*, a full-day participatory workshop exploring how computational interfaces scaffold collaborative flavor creation of a popular drink—boba tea. We will use an LLM-supported *Flavor*

Wiki system to record participants’ sensory impressions, personal and cultural associations, flavor interpretations, and finally guide them to iteratively create customized drinks.

1 Introduction

Flavor is a multisensory construct that is an integration of multiple senses and sociocultural, mnemonic dimensions [3, 22]. Human-Food Interaction (HFI) explores this field as a broad design space for computational approaches [12]. Previous works have modeled flavor data using recipes and chemical compounds, such as [1], FlavorGraph [21], FlavorDB [8], and, more recently, LLM agents for flavor hypotheses [10]. Researchers also have explored multimodal flavor presentation, which highlights the potential to computationally create new experiences. For example, AR pseudo-gustatory displays [18, 19] and chemical taste and smell interfaces [4, 6, 13].

However, current technology for reconstructing flavor experience still has clear limits. For example, some details, such as smell

Permission to make digital or hard copies of all or part of this work for personal or classroom use is granted without fee provided that copies are not made or distributed for profit or commercial advantage and that copies bear this notice and the full citation on the first page. Copyrights for components of this work owned by others than the author(s) must be honored. Abstracting with credit is permitted. To copy otherwise, or republish, to post on servers or to redistribute to lists, requires prior specific permission and/or a fee. Request permissions from permissions@acm.org.
UIST '26, Detroit, MI, USA

© 2026 Copyright held by the owner/author(s). Publication rights licensed to ACM.
ACM ISBN 978-x-xxxx-xxxx-x/YYYY/MM
<https://doi.org/10.1145/nnnnnnn.nnnnnnn>

and flavor, are hard to verbalize because of limited olfactory vocabulary and cross-linguistic variation [14] to evolve from the existing flavor profile taxonomy [7, 15]. This leaves a room for data-driven systems to understand how flavor is interpreted subjectively, especially situated accounts shaped by life experiences and cultural contexts posing technical problems for HFI research[16]. There are prior tools helping individuals externalize ephemeral taste. such as the FlavourFrame using layered visual–textual canvases for mindful tasting [2], and FlavorDesigner supporting structured capture of multisensory experiences and personalized flavor-memory cues [9].

In this workshop, we extend from individual to collaborative, co-editing accounts for better capturing social effects. In brief, we focus on a key question: *how might computational interfaces help communities articulate, share, and iteratively refine their personal interpretations of taste?*

We choose boba tea as our medium to explore the question for both its cultural content and functional features. Boba tea carries a long history of cultural adaptation from the movement of tea and sugar through colonial trade across continents to thriving industry adapting into different culture zone [11, 20]. Beyond this historical context, much of this popularity also stems from standardized menus paired with extensive freedom to customize—sweetness, ice level, milk type, tea base, and toppings—positioning boba tea as a lightweight form of participatory food design.

2 Participants and Recruitment

The workshop aims to attract 30 participants from HFI, multisensory human-computer interactions, human-AI co-creation, and culinary practice; no culinary skills are required. We recruit participants with diversity of background using short open-ended questions: (1) language, (2) cultural region, and (3) professional backgrounds, and select by balancing groups with these aspects.

3 Workshop Activities

Time	Block	Activity
09:30–10:45	Opening & Talks	Welcome, logistics, two invited talks
10:45–11:05	Technique	Boba-making demo
11:05–12:20	Task 1	Flavor entity creation + system intro
12:20–14:00	Task 2 (lunch)	Ingredient-led boba design; document in wiki
14:00–15:30	Task 3	Memory- or theme-driven boba; document in wiki
15:30–16:15	Share	Present drinks, open tasting & wiki annotation
16:15–17:30	Discussion & Close	Group reflection, research directions, closing

Table 1: The workshop schedule. Tasks follow an iterative cycle; regroup after each round.

Bobapedia is a full-day workshop, moving from individual sensory documentation to group comparison, concept design, and iteration.

Task 1 — Flavor Entity Creation

Participants taste and smell assigned ingredients; the system assists them in describing sensory attributes, associations, and cultural references—eliciting follow-ups when utterances collapse to generic terms and custom visualization.(Figure 1b) This session also breaks the ice while participants practice articulating flavor.

Task 2 — Ingredient-Led Design

Groups select an anchor ingredient from the shared pool, design a boba tea around it, and record rationale and expected sensory profile in the wiki before preparing and tasting (lunch at the same time so dining itself becomes a part of the documentation context).

Task 3 — Memory- and Theme-Driven Design

Groups reshuffle, identify a shared memory or theme, and translate it into a boba concept using earlier flavor entries as building blocks, tracing ingredients back to that theme in the wiki.

Sharing, Discussion, and Close

Groups present their drinks alongside Flavor Wiki entries. Participants then circulate to taste one another’s creations and add annotations to shared wiki entries beyond their own group’s contributions.

An extended discussion addresses tensions between participants, LLM normalization and idiosyncratic expressions. Throughout this phase, a shared dashboard displays aggregated flavor visualizations and co-created archives so participants can compare ideas and design decisions across teams (Figure 1c).

4 Health, Dietary, and Venue Considerations

We will ensure ingredients fit common allergen, religious (e.g., halal), vegan, and low-sugar requirements, or provide alternatives when needed, and use small servings with palate cleansers to limit palate fatigue. For venue logistics, we will ensure preparation that do not require extensive on-site work or other complex procedures.

5 Organizers

The organizing team brings together expertise in flavor and chemical interfaces, multisensory, multimodal AI, and participatory food design. Brooks and Narumi have previously hosted workshops on smell, taste, and human–food interaction [5, 17].

6 Expected Outcomes and Future Work

The workshop will produce a collaboratively edited Flavor Wiki—annotated ingredient entries and boba tea concepts capturing not only recipes, but the sensory, cultural, and narrative reasoning from individuals.

We will publish these materials on the workshop website. More broadly, the workshop aims to build a research agenda that connects data-driven flavor models with embodied experience and supports interpretive diversity. We also explore how computational systems can help communities co-edit and extend everyday food knowledge. This agenda can inform future work across HFI, gastronomy, multisensory interaction, and human–AI creativity support.

References

- [1] Yong-Yeol Ahn, Sebastian E. Ahnert, James P. Bagrow, and Albert-László Barabási. [n.d.]. Flavor Network and the Principles of Food Pairing. 1, 1 ([n.d.]), 196. <https://doi.org/10.1038/srep00196>
- [2] Lauryn Anderson, Lora Oehlberg, and Wesley Willett. [n.d.]. FlavourFrame: Visualizing Tasting Experiences. In *Adjunct Proceedings of the 36th Annual ACM Symposium on User Interface Software and Technology* (San Francisco CA USA, 2023-10-29). ACM, 1–3. <https://doi.org/10.1145/3586182.3616640>
- [3] Malika Auvray and Charles Spence. 2008. The multisensory perception of flavor. *Consciousness and cognition* 17, 3 (2008), 1016–1031.
- [4] Jas Brooks, Noor Amin, and Pedro Lopes. 2023. Taste Retargeting via Chemical Taste Modulators. In *Proceedings of the 36th Annual ACM Symposium on User Interface Software and Technology*. ACM. <https://doi.org/10.1145/3586183.3606818>
- [5] Jas Brooks, Pedro Lopes, Judith Amores, Emanuela Maggioni, Haruka Matsukura, Marianna Obrist, Roshan Lalintha Peiris, and Nimesha Ranasinghe. 2021. Smell, Taste, and Temperature Interfaces. In *Extended Abstracts of the 2021 CHI Conference on Human Factors in Computing Systems*. ACM, 1–4. <https://doi.org/10.1145/3411763.3441317>
- [6] Jas Brooks, Steven Nagels, and Pedro Lopes. 2020. Trigeminal-based Temperature Illusions. In *Proceedings of the 2020 CHI Conference on Human Factors in Computing Systems*. ACM, 1–12. <https://doi.org/10.1145/3313831.3376806>
- [7] SE Cairncross and LB Sjostrom. 2004. Flavor profiles: a new approach to flavor problems. *Descriptive sensory analysis in practice* (2004), 15–22.
- [8] Neelansh Garg, Apuroop Sethupathy, Rudraksh Tuwani, N. K. Rakhi, Shubham Dokania, Arvind Iyer, Ayushi Gupta, Shubhra Agrawal, Navjot Singh, Shubham Shukla, Kriti Kathuria, Rahul Badhwar, Rakesh Kanji, Anupam Jain, Avneet Kaur, Rashmi Nagpal, and Ganesh Bagler. [n.d.]. FlavorDB: A Database of Flavor Molecules. 46, D1 ([n.d.]), D1210–D1216. <https://doi.org/10.1093/nar/gkx957>
- [9] Tom Gayler, Corina Sas, and Vaiva Kalnikaitė. [n.d.]. FlavorDesigner App: Capturing Multisensory Experiences and Crafting Personalized Flavors for Cueing Their Recall. In *Designing Interactive Systems Conference* (Virtual Event Australia, 2022-06-13). ACM, 1441–1456. <https://doi.org/10.1145/3532106.3533474>
- [10] Tenghao Huang, Donghee Lee, John Sweeney, Jiatong Shi, Emily Steliotes, Matthew Lange, Jonathan May, and Muhao Chen. [n.d.]. FoodPuzzle: Developing Large Language Model Agents as Flavor Scientists. ([n.d.]). <https://doi.org/10.48550/arXiv.2409.12832> arXiv:2409.12832
- [11] Po-Yi Hung. 2022. Food Nationalism Beyond Tradition: Bubble Tea and the Construction of Taiwanese Identity. *International Journal of Gastronomy and Food Science* 29 (2022), 100567. <https://doi.org/10.1016/j.ijgfs.2022.100567>
- [12] Rohit Ashok Khot and Florian “Floyd” Mueller. [n.d.]. Human-Food Interaction. 12, 4 ([n.d.]), 238–415. <https://doi.org/10.1561/11000000074>
- [13] Jasmine Lu, Ziwei Liu, Jas Brooks, and Pedro Lopes. 2021. Chemical Haptics: Rendering Haptic Sensations via Topical Stimulants. In *The 34th Annual ACM Symposium on User Interface Software and Technology*. ACM, 239–257. <https://doi.org/10.1145/3472749.3474747>
- [14] Asifa Majid and Niclas Burenhult. 2014. Odors Are Expressible in Language, as Long as You Speak the Right Language. *Cognition* 130, 2 (2014), 266–270. <https://doi.org/10.1016/j.cognition.2013.11.004>
- [15] Morten C. Meilgaard. 1975. Flavor chemistry of beer. I. Flavor interaction between principal volatiles. <https://api.semanticscholar.org/CorpusID:93560509>
- [16] Florian “Floyd” Mueller, Marianna Obrist, Ferran Altarriba Bertran, Nithya Makam, Seung Ah Kim, Christopher Dawes, Patrizia Marti, Maurizio Mancini, Eleonora Ceccaldi, Nandini Pasumarthy, S. Claire, K. Seo Jung, Jialin Deng, Jürgen Steimle, N. Krasteva, M. Schwalk, Harald Reiterer, H. Wang, and Yan Wang. [n.d.]. Grand Challenges in Human-Food Interaction. 183 ([n.d.]), 103197. <https://doi.org/10.1016/j.ijhcs.2023.103197>
- [17] Takuji Narumi. 2016. Multi-sensorial Virtual Reality and Augmented Human Food Interaction. In *Proceedings of the 1st Workshop on Multi-sensorial Approaches to Human-Food Interaction*. ACM, 1–6. <https://doi.org/10.1145/3007577.3007587>
- [18] Takuji Narumi, Takashi Kajinami, Shinya Nishizaka, Tomohiro Tanikawa, and Michitaka Hirose. 2011. Pseudo-gustatory Display System Based on Cross-modal Integration of Vision, Olfaction and Gustation. In *2011 IEEE Virtual Reality Conference*. IEEE, 127–130. <https://doi.org/10.1109/vr.2011.5759450>
- [19] Takuji Narumi, Munehiko Sato, Tomohiro Tanikawa, and Michitaka Hirose. 2010. Evaluating Cross-sensory Perception of Superimposing Virtual Color onto Real Drink. In *Proceedings of the 1st Augmented Human International Conference*. ACM, 1–6. <https://doi.org/10.1145/1785455.1785473>
- [20] National Geographic. 2024. Origins and Cultural Impact of Boba Tea, Taiwan’s Iconic Drink. <https://www.nationalgeographic.com/travel/article/what-is-boba-bubble-tea-taiwan> Accessed 2026-05-29.
- [21] Donghyeon Park, Keonwoo Kim, Seoyoon Kim, Michael Spranger, and Jaewoo Kang. [n.d.]. FlavorGraph: A Large-Scale Food-Chemical Graph for Generating Food Representations and Recommending Food Pairings. 11, 1 ([n.d.]), 931. <https://doi.org/10.1038/s41598-020-79422-8>
- [22] Dana M Small and John Prescott. 2005. Odor/taste integration and the perception of flavor. *Experimental brain research* 166, 3 (2005), 345–357.